2025 Summer Qual (Master)

$\boxed{1}$.

(a) $\mathbb{E}[y^T A y] = \mathbb{E}[\mathrm{tr}(y^T A y)]$

$$= \mathbb{E}[\mathrm{tr}(A y y^T)]$$

$$= \mathrm{tr}[\mathbb{E}[A y y^T]]$$

$$= \mathrm{tr}(A(\Sigma + \mu\mu^T))$$

$$= \mathrm{tr}(A\Sigma) + \mu^T A \mu$$

$\blacksquare$

(b) $\mathrm{cov}(y, y^T A y) = \mathbb{E}[y - \mathbb{E}[y]][y^T A y - \mathbb{E}[y^T A y]]$

$$= \mathbb{E}[y - \mu][y^T A y - \mathrm{tr}(A\Sigma) - \mu^T A \mu]$$

$$= \mathbb{E}[y - \mu][(y-\mu)^T A (y-\mu) + 2(y-\mu)A\mu - \mathrm{tr}(A\Sigma)]$$

$$= \underbrace{\mathbb{E}[y-\mu][(y-\mu)^T A (y-\mu)]}_{3\text{차 moment}} + \underbrace{2\mathbb{E}[y-\mu][(y-\mu)A\mu]}_{2\text{차 moment}} - \underbrace{\mathbb{E}[y-\mu]\mathrm{tr}(A\Sigma)}_{1\text{차 moment}}$$

$$= 0 + 2\Sigma A\mu - 0$$

$$= 2\Sigma A \mu$$

$\blacksquare$

(c) $M_{y^TAy}(t) = |I - 2tA\Sigma|^{-1/2} \exp\left(-\frac{1}{2}\mu^T[I - (I-2tA\Sigma)^{-1}]\Sigma^{-1}\mu\right)$

$C(t) := \log M_{y^TAy}(t)$

$= \frac{-1}{2}\log|C| - \frac{1}{2}\mu^T(I - C^{-1})\Sigma^{-1}\mu \quad , C := I - 2tA\Sigma$

$C'(t) = -\frac{1}{2}tr\left(C^{-1}\frac{\partial C}{\partial t}\right) - \frac{1}{2}\mu^T\left(-\left(-C^{-1}\frac{\partial C}{\partial t}C^{-1}\right)\right)\Sigma^{-1}\mu$

$= tr(C^{-1}A\Sigma) - 2\mu^TC^{-1}A\Sigma C^{-1}\Sigma^{-1}\mu$

$C''(t) = tr\left(-C^{-1}\frac{\partial C}{\partial t}C^{-1}A\Sigma\right) - 2\mu^T\left(-C^{-1}\frac{\partial C}{\partial t}C^{-1}\right)A\Sigma C^{-1}\Sigma^{-1}\mu$

$- 2\mu^TC^{-1}A\Sigma\left(-C^{-1}\frac{\partial C}{\partial t}C^{-1}\right)\Sigma^{-1}\mu$

$C\big|_{t=0} = I, \quad \frac{\partial C}{\partial t}\Big|_{t=0} = -2A\Sigma$

$C''(0) = 2\,tr(A\Sigma A\Sigma) + 4\mu^TA\Sigma A\mu$

$= Var(y^TAy)$

$\boxtimes$

2.

(a) $\hat{\beta} = (X^T X)^{-1} X^T y$, $\hat{\varepsilon} = (I - H_X) y$

$E[\hat{\varepsilon}] = (I - H_X) X\beta = 0$

(b) $\hat{\varepsilon}^T \hat{y} = y^T (I - H_X) H_X y = y^T (H_X - H_X) y = 0$

$\hat{\varepsilon}^T X = y^T (I - H_X) X = y^T (X - X) = 0$

(c) $\mathrm{cov}(\hat{\varepsilon}, \hat{y}) = \mathrm{cov}((I - H_X) y , H_X y)$

$= (I - H_X) \Sigma H_X \neq 0 \quad \because \Sigma$

(d) $\hat{\varepsilon}^T 1 = y^T (I - H_X) 1 = 0$

$\Rightarrow H_X 1 = 1$ or $1 \in C(X)$

$\boxtimes$